

The International Journal of Aerospace Psychology

ISSN: 2472-1840 (Print) 2472-1832 (Online) Journal homepage: www.tandfonline.com/journals/hiap21

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To cite this article: Lakshmi Vempati, Sabrina Woods & Robert C. Solano (2023) Qualitative Analysis of General Aviation Pilots' Aviation Safety Reporting System Incident Narratives Using the Human Factors Analysis and Classification System, The International Journal of Aerospace Psychology, 33:3, 182-196, DOI: 10.1080/24721840.2023.2232387

To link to this article: https://doi.org/10.1080/24721840.2023.2232387

Published online: 20 Jul 2023.

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THE INTERNATIONAL JOURNAL OF AEROSPACE PSYCHOLOGY
2023, VOL. 33, NO. 3, 182–196
https://doi.org/10.1080/24721840.2023.2232387

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Qualitative Analysis of General Aviation Pilots' Aviation Safety Reporting System Incident Narratives Using the Human Factors Analysis and Classification System

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ABSTRACT

NASA's Aviation Safety Reporting System (ASRS) gathers data from voluntary aviation safety reports for nonfatal accidents and incidents. These reports are a valuable resource in identifying unsafe occurrences and hazardous situations in the aviation industry. A tool known as the Human Factors Analysis and Classification System (HFACS), initially developed for military use, has proven useful for root cause analysis of human causes in aviation accidents and incidents, including in commercial contexts. This research study utilizes HFACS to classify the ASRS safety reports to identify key indicators in General Aviation (GA) non-fatal accidents and incidents. Qualitative data analysis reveals relationships between incidents and causal factors, indicating 53% of incidents to be perceptual errors, 32% decision errors, and 26% skill-based errors. Prevalent preconditions for these errors include crew resource management (53%), physical environment (32%), and technological environment (16%). The study aligns with previous research findings, suggesting that crew resource management is a common factor in both GA incidents and commercial and military aviation. While GA pilots may make different types of errors, such discrepancies are not prevalent in the voluntarily reported data.

Introduction

The NASA Aviation Safety Reporting System (ASRS) was implemented in 1975 and was originally designed to be a confidential system used by anyone functioning in the National Airspace System (NAS; Billings et al., 1976). The computer-based system collects and stores data to be used for analysis in research, regulatory, operational, and safety advocacy capacities (Billings et al., 1976). Special consideration is given to the reporting system's voluntary nature as it affords users a certain level of anonymity and potential immunity in reporting, while providing a more expansive mechanism for data collection. The reports that are submitted to ASRS describe both unsafe occurrences and hazardous situations. In particular, ASRS focuses on the quality of human performance in the NAS and analyses incidents (rather than accidents) as a form of leading indicator data for aviation safety (Billings, 1998; National Aeronautics and Space Administration [NASA], 2015).

The Human Factors Analysis and Classification System (HFACS), a human error framework developed by Douglas Wiegmann and Scott Shappell, was designed to be a tool to

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provide root cause analysis on the human causes of aviation accidents. HFACS was originally based upon James Reason's *Swiss Cheese* model for latent and active failures but was then expanded to address external and internal human factors throughout the accident causation chain (Salmon, 2011; Shappell & Wiegmann, 2001). The HFACS framework establishes an additional 19 causal categories within Reason's (1990) original four levels of human failure (Figure 1).

Although many safety measures and accident investigation procedures have ensured that air transportation remains the safest mode of travel (Chambers, 2011; Savage, 2013), for Part 91 general aviation (GA), the yearly fatal mishap rate has remained rather stagnant over the previous five years leading to 2015 (Figure 2).

Further, while commercial air transport has made great strides in reducing the fatal accident rate to nearly zero, GA has hovered between 1.1 and 1.2 mishaps for every 100,000 hours of flight (Bertorelli, 2015). Germane to any examination of GA accident or incident are the unsafe acts of aircrew, therefore those acts bear the most scrutiny when conducting an investigation (Wiegmann and Shappell, 2001).

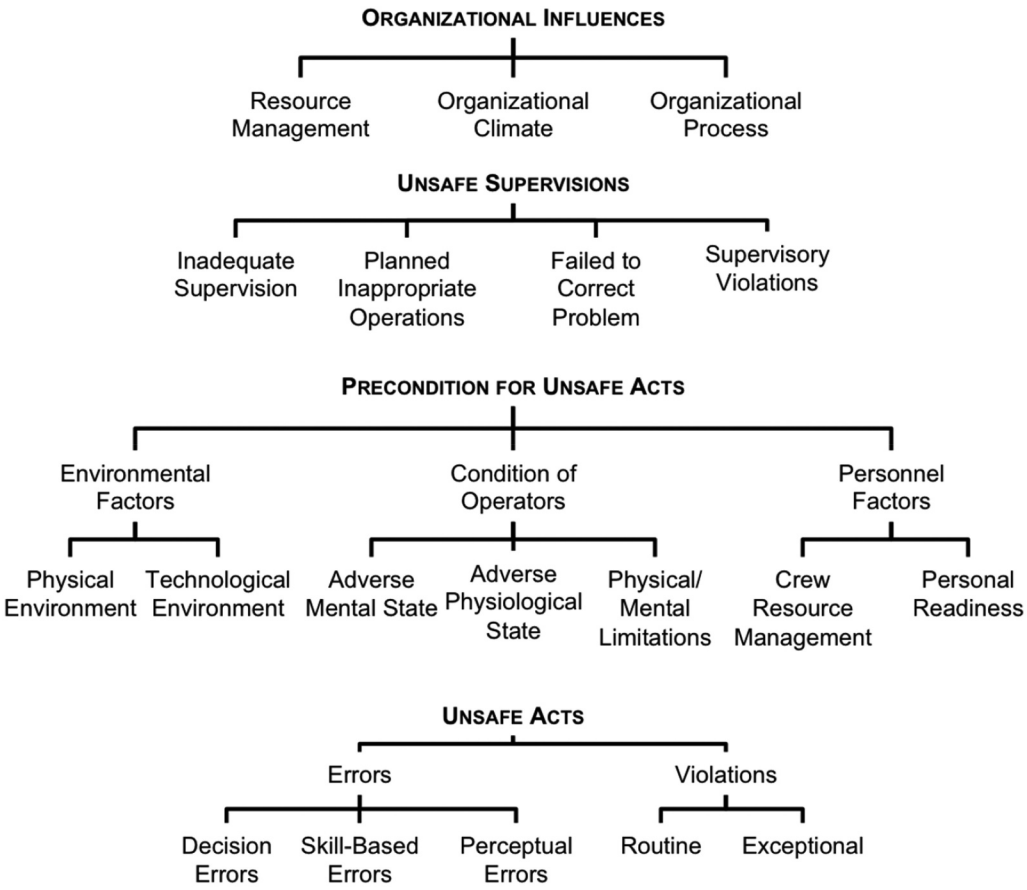


Figure 1. HFACS framework. reprinted from human error and commercial aviation accidents: An analysis using the human factors analysis and Classification System Shappell, S., Detwiler, C., Holcomb, K., Hackworth, C., Boquet, A., & Wiegmann, D. A. (2007). Human factors, 49(2), 227–242.

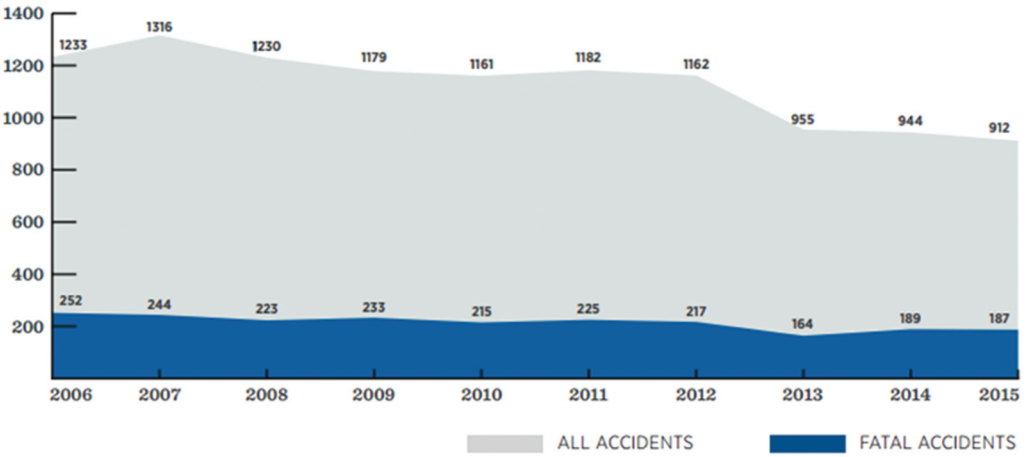


Figure 2. Noncommercial fixed-wing general aviation mishaps for 2006 to 2015. Reprinted from 27th Joseph T. Nall report-general aviation accidents in 2015. Aircraft Owners and Pilots Association (2018) https://www.aopa.org/-/media/Files/AOPA/Home/Training-and-Safety/Nall-Report/27thNallReport2018.pdf.

Previously, safety mitigation strategies have typically been the result of an investigation succeeding a mishap (Halford et al., 2015). The aviation safety paradigm is shifting, however, to becoming less reactive by implementing more proactive tools, procedures and initiatives that encourage safety (Halford et al., 2015). In 2001, Wiegmann and Shappell applied their tool to general aviation in order to identify general trends in the types of human factors issues and aircrew errors that contribute to civil aviation accidents. Their analysis was largely based on post-accident (i.e., lagging) data obtained from the National Transportation Safety Board (NTSB) and the FAA's National Aviation Safety Data Analysis Center (NASDAC) (Wiegmann & Shappell, 2001).

Aviation accidents rarely happen in isolation (Lenné et al., 2008; Shappell & Wiegmann, 2017; Wiegmann & Shappell, 2001). Rather, a mishap is typically the direct or indirect result of a series of events that Wiegmann and Shappell (2001) believe culminate in an unsafe act a person (usually the pilot) has taken. Humans, by nature, make mistakes and, in over 70% of aviation mishaps, the human is found to be a primary causal factor (Salas et al., 2010; Shappell & Wiegmann, 2017). Numerous studies have been accomplished on accident causation and post-mishap analysis to better understand the genesis of human error as it relates to the mishap (Wiegmann & Shappell, 2001). In particular, three prominent mishaps have shaped the way accidents are investigated, and have highlighted the importance of understanding the human role in accident causation are summarized here:

DHL Flight 611 and Bashkirian Flight 2937 collided in 2002 over the skies of Überlingen, Germany killing all on board both aircraft. The two aircraft entering Swiss airspace were accidentally set on a collision course by the one controller working the airspace at the time. The traffic collision avoidance system (TCAS) onboard both aircraft attempted to deconflict the flight paths, however the crew of the Bashkirian flight elected to ignore it and instead followed the guidance of the controller, who had no knowledge of the TCAS alerting. While the pivotal accident typically serves as a case study for systemic issues in air traffic control

(Bennett, 2004), the decisions of the Bashkirian crew played a significant role in the accident causation chain. This mishap can also be applied to all four levels of the HFACS model.

On July 16, 1999, John F. Kennedy Jr. took off in a newly purchased Piper *Saratoga* heading for Martha's Vineyard with his wife and sister-in-law. He never made it as the *Saratoga* crashed into the Atlantic Ocean off the coast of Massachusetts. All three people onboard were killed. The subsequent NTSB investigation determined Kennedy, who did not hold an instrument rating and was unfamiliar with the aircraft, became spatially disorientated while attempting to descend at night and over water (National Transportation Safety Board [NTSB], 2000). In summary, the reason for the mishap was almost entirely pilot error. The HFACS model application for these accidents was *unsafe acts (error)* and *precondition for unsafe acts*.

More recently, Air France Flight 447 crashed into the Atlantic Ocean on June 1, 2009 killing all on board. Icing on the airspeed sensors on the plane's wings and fuselage set off the chain of events that ultimately led to the crash (Bureau of Enquiry and Analysis for Civil Aviation Safety [BEA], 2012). Investigators determined that those technical failures were exacerbated by several mistakes made by inadequately trained pilots who made several *perceptual, skill*, and *decision-based errors* that eventually put the aircraft in an aerodynamic stall from which they were unable to recover (BEA, 2012).

Aviation mishap and safety investigators have worked to grasp the psychological aspects underlying human error. In each example, the post mishap analyses have affected change in technology, procedures, weather data collection, and education and safety outreach initiatives. These changes are still *after the fact*, however; and based on latent data. The next step is to apply these models to incidents (situations that have not yet manifested into accidents) in order to identify the exact nature of human error and hopefully assist in the generation of intervention programs. A quick search of applicable case studies, journal articles, and analyses from 2008 to 2015 returned very few dedicated to building a framework and applying human factors analysis on incidents that were resolved without issue, versus accidents.

In 2008, Lenné, Ashby, and Fitzharris applied the HFACS model to general aviation mishaps in Australia. They determined that aircrew were more likely to commit skill-based and decision errors when the precondition for an unsafe act was personal readiness, physical and mental limitations, or adverse mental state (Lenné et al., 2008). Violations were associated with issues in crew resource management (CRM). While an important study in developing correlations between factors and behaviors, this study was still based on post-accident data.

Bearman et al. (2010) used ASRS reports to explore how coordinated decision-making degrades between pilots and air traffic controllers, and the way those situations are ultimately resolved. In their study, Bearman et al. (2010), highlight the fact that pilots and controllers have very different roles and responsibilities; however, they must work together to make coordinated decisions. If the two groups do not have the same goals, dissonance occurs and that coordination breaks down (Bearman et al., 2010). The researchers then tried to determine the pattern the breakdown follows by identifying patterns in ASRS narratives. For this study, ASRS facilitated the initial categorization of reports for use in trending and research.

The Joseph T. Nall report is a bi-yearly product produced by the Aircraft Owners and Pilots Association (AOPA) Air Safety Institute (ASI). It reviews GA NTSB accident and

incident reports for which reasonably complete data are available and provides an analysis of the various accidents, statistics and trends such as type of operation, flight conditions, and major causes. The report defines accidents as an incident where, "as a result of the operation of an aircraft, any person (occupant or non-occupant) who receives a fatal or serious injury or any aircraft receives substantial damage" (Aircraft Owners and Pilots Association [AOPA], 2015, p. 51). The reports aid in developing the taxonomies necessary to organize narrative data in order to establish themes and determine relationships for HFACS classification.

Last, and probably the most beneficial study found in the literature review is by Olsen (2011) who researched and developed a coding taxonomy based on air traffic control incident narratives. She noted the accuracy of the analyses, which are vital to the development of accident prevention measures, are entirely dependent on the subject matter experts' ability to come to a consensus on what they feel is the root cause of each incident (Olsen, 2011). While this study did not add any relevant insight on human error in incident reporting, it did aid in how the researchers formatted their own analysis of ASRS data prior to coding for NVivo.

The relative absence of HFACS (or any other framework) application to incident data practices is telling. The results show a need for additional studies to be focused on incident report data with the same level of intensity afforded to post accident data. Even the Federal Aviation Administration (FAA) recognizes the need to pursue input through reports from employees, industry advocates, and other users of the National Airspace System in order "to recognize the many opportunities exist to stop an accident" (FAA, 2016, para 1).

The Cessna 172 *Skyhawk* was chosen as a basis for study. The *Skyhawk* is a four-seat, single-engine, high-wing aircraft (Mammone, 2013). The aircraft was first received from the production line in 1955, is still in production, and to date more Cessna 172s have been built than any other aircraft (Mammone, 2013). Twenty five percent of the Cessna aircraft currently registered in the FAA Aircraft Registry comprise of C172s. It is historically one of the most successful planes produced for the GA industry and therefore was selected as the representative subject aircraft for the study. ASRS data from 2015, and reported by Cessna 172 pilots, was pulled rendering over 1000 results.

In summary, the Wiegmann and Shappell (2005) study showcased that within GA accidents, literally thousands of unsafe acts are committed by pilots, which in turn suggests there are multitudes of unique ways to crash an airplane. The study continued by demonstrating accidents that may appear unique at the surface level can still be grouped into a structure by which significant themes, trends and relationships can be deduced (Wiegmann & Shappell, 2005). It was Wiegmann and Shappell's (2005) hope that such data "would result in more data-driven intervention efforts being developed and implemented" (p. 5). While Wiegmann and Shappell analyzed post-accident data, no such analysis exists for aircraft incidents, which do not exceed the threshold for an accident investigation.

Current Study

The purpose of the current study was to continue where Wiegmann and Shappell left off by assessing the utility of the HFACS framework as an error analysis and classification tool for

information acquired through the ASRS system. Using data submitted in 2015 by Cessna 172 pilots, the research team used the leading indicator data found within general aviation ASRS reported incident narratives and coded them using the HFACS model. By doing this the team hoped to provide deeper insight into the types of human errors and preconditions for those errors in general aviation incidents. The study was guided by the following research question:

1. Using the HFACS framework, what are the frequencies of the different types of human error and preconditions for those errors in general aviation incidents as reported through ASRS?

Research Methodology

A summary review of the last three Nall reports (23rd, 24th, and 25th, respectively) indicated human error, often exacerbated by either environmental, technological, or human factors, remain a key finding of significance in post-mishap analyses (Aircraft Owners and Pilots Association [AOPA], 2014, 2015, 2016). The categorical grouping of these findings is consistent with the framework of the HFACS model. Therefore, narratives from 2015 pertaining to Cessna 172 incidents were queried directly from the NASA ASRS database. This resulted in a set of 48 narratives that highlighted human elements as having a role in the incident. In order to further refine the data and keep the focus on human error, narratives involving mechanical failure of the aircraft and those reported by individuals other than the primary or secondary operator of the aircraft were eliminated from the study. The final set of narratives comprised of 38 incidents, limited to those that occurred during final approach or landing phase under visual meteorological conditions.

Once the final narratives were selected, the team coded each based on the HFACS model. The upper two tiers, *organizational influences* and *unsafe supervision*, are categories typically indicative of an established body of people who are bound together by a specific purpose such as a business or an initiative (e.g., military) and who have a hierarchal structure of leadership. Upon further research, it was determined that the general aviation activities of interest for this study do not meet the definition of an organization or formalized hierarchy therefore only the lower two tiers, *unsafe acts* and *precondition for unsafe acts*, were applied.

Each member of the team reviewed each of the 38 narratives and classified each of them following the HFACS model. After considerable discussion, any difference in the classification of the narratives was resolved and the final choices were coded using the HFACS model. Once each narrative was coded, NVivo, a qualitative data analysis tool was used to classify, sort, and examine relationships within the data. The researchers coded the narratives via interactive coding and in accordance with the HFACS framework.

Results

A classification analysis identified the primary unsafe act and the precondition for each of the ASRS reports. The query of the code is shown in Table 1.

Perceptual errors were the most frequent unsafe act at 53%, followed by decision errors (32%) and then skill-based errors with 26%. A common perceptual error was a pilot failing to see another aircraft, such as " . . . the other traffic was near and below we did not have a chance to see them on until we turned final" (report number 1252857). While another

Table 1. Percentage of accidents associated with each HFACS category.

HFACS Category	ASRS Part 91[a] Reports per Category	Part 121 & 135[b] Reports per Category
Preconditions for Unsafe Act		
Adverse Mental State	5% (2)	14% (16)
Adverse Physiological State	0	2% (2)
Physical or Mental Limitations	5% (2)	11% (13)
Physical Environment	32% (12)	n/a
Technological Environment	16% (5)	n/a
Crew Resource Management	53% (17)	29% (35)
Personal Readiness	3% (1)	0%
Unsafe Acts		
Decision Errors	32% (12)	29% (34)
Perceptual Errors	53% (17)	14% (17)
Skill-based Errors	26% (10)	61% (72)
Exceptional or Routine Violation	0	27% (32)

The numbers in the table are the percentages of accidents that involved the indicated HFACS category. The number in parenthesis is the accident frequency. [a] The Part 91 data is based upon qualitative analysis of ASRS reports (2016). [b] The Parts 121 and 135 data is based upon findings of Wiegmann and Shappell (2001). Physical environment and technological environment were not recognized HFACS categories at the time of Wiegmann and Shappell's research.

type of perceptual error involved failures to accurately identify ground references, such as an aircraft intending to land at an airport but instead, "setting up to land on a street" (report number 1248819).

The most common precondition for unsafe acts was crew resource management (53%), followed by physical environment (32%) and then technological environment (16%). It is important to note that although crew resource management most commonly refers to interactions between crew members aboard the same aircraft, HFACS defines this category more broadly. According to HFACS, Inc. (2012), this category includes a "variety of communication, coordination, and teamwork issues that impact performance" (p. 1). In the ASRS reports, most of the incidents that fell within this category involved poor communication, or complete lack of communication, between multiple aircraft or air traffic controllers. An example is an incident that reported, "I made a radio call on CTAF asking who was on the runway, but no one responded. The lights then lifted from the ground directly into my path, it was a B747" (report number 1274125)

The ASRS reports are voluntarily reported; as such, it was not unusual for the reporter to perceive and therefore indicate another aircraft or air traffic controller to be at fault for the incident. This is not to be a slight against the people who willingly volunteer their experiences for the sake of aiding in the data collection process. People are fundamentally driven to seek causal explanations for how and why things happen (Lagnado & Channon, 2008). However, attribution research has determined that intention and foresight influence blame and have a distorting effect on causal judgments (Alicke, 2000; Lagnado & Channon, 2008). This distortion is based on the human presumption that someone is at fault for every unfavorable situation and is exacerbated by an inability to see how one's own actions and decisions have affected the outcome of said situation (Alicke, 2000; Lagnado & Channon, 2008).

Because the ASRS reports only provide one perspective of each incident, the subject matter experts conducting the analysis accepted the reports at face value. For example, if a pilot reported that another aircraft was not reporting their actions on a radio, then the researchers assumed that the reporting pilot was listening to the proper radio

frequency and that the pilot's radios were operating properly. Therefore, the researchers would conclude that the other pilot was at fault for poor communication and coordination. Although it is possible that other unidentified HFACS categories may have been a precondition for the unsafe act, the researchers assumed that each report was a reasonable account of what took place. Deeper analysis would require a full incident investigation with multiple interviews, which is beyond the scope of this research.

The second most common precondition for an unsafe act was the physical environment. A word frequency query of these particular incidents indicated that runway, landing, airport, winds, approach, and light were some of the most common words as depicted in Table 2. The table counts provide greater insight into what types of physical environments are most often associated with errors. Future research can build upon these findings to develop higher fidelity codes or potential subcategories for HFACS preconditions. These findings show that most environmental preconditions to unsafe acts involved aircraft flying near airports, were typically attempting to land, and either experienced poor lighting conditions or strong winds. While most environmental conditions involved lighting or winds, there was one outlier which involved a ship on the waterway on the approach end of a runway which blocked the aircraft's approach angle.

When compared to Part 121 and 135 as shown in Table 1, Part 91 incidents had similar results for the preconditions to unsafe acts. Both the Wiegmann and Shappell (2001) report and the ASRS incidents (2016) identified that crew resource management was the leading HFACS precondition to unsafe acts. It is difficult to compare the second most common precondition since physical and technological environment factors were recently added HFACS categories and not included in the 2001 report.

The types of unsafe acts identified for ASRS Part 91 incidents were drastically different from the Part 121 and 135 accidents. While skill-based errors and decision-based errors were the most common incidents for Parts 121 and 135, perception errors were the most common for the ASRS reports. This follows with the logic that individuals are less likely to voluntarily report their own shortcomings. One possible hypothesis is that individuals are more comfortable to report that they did not see another aircraft, rather than reporting that an incident occurred due to their poor decisions or lack of skill.

Table 2. Top twelve words for environmental factors preconditions for unsafe acts.

Word	Weighted Percentage (%)
runway	2.96
aircraft	1.55
landing	1.18
right	1.18
airport	1
approach	0.96
left	0.96
pattern	0.86
around	0.73
ctaf	0.73
tower	0.73
winds	0.73

Cross Component Analysis

Graphical analysis to determine the relationships between unsafe acts and the preconditions for those unsafe acts, as well as the relationship between the most common unsafe act, perceptual errors, and the most common precondition for unsafe acts, crew resource management did not provide any clear relationships. While a graphical map of the nodes did not provide significant results, a node cluster analysis did provide further insight. The node cluster analysis provides a tree diagram that links similar nodes which are most commonly coded together. The node cluster analysis provided details that would otherwise be unapparent through other graphical means. As shown in Figure 3, the cluster analysis showed that crew resource management and perceptional errors pair with each other more than other HFACS categories. The other significant HFACS pair was the physical environment and skill-based errors indicating that these categories are also most commonly paired together.

Next, an analysis of the classification was conducted using a classification sheet to determine if time of day, daylight hours or contributing factors as contained within each report provided any common patterns. It should be mentioned that the incidents presented in this analysis are visual meteorological incidents experienced in a Cessna 172 aircraft. Each reported incident contains one or more contributory factors and a classification sheet was designed using the provided primary contributory factors that were revised to follow a common name convention to facilitate analysis. The relationship of errors and preconditions by the time of day and the daylight hours when the reported incidents occurred is shown in Figures 4 and 5 respectively.

As can be observed from Figure 4, majority of the events occurred during daylight hours between 0600 and 2400 Zulu time. There were fewer reported incidents during nighttime and as already observed in the HFACS classification, the primary preconditions for the incidents can be attributed to crew resource management or physical environment.

Next, the relationships between the primary contributing factor as contained in the incident report and the classified HFACS precondition was explored as shown in Figure 6.

As expected, almost 50% of the incidents were when conflict was the primary contributing factor and classified as crew resource management precondition due to lack of proper

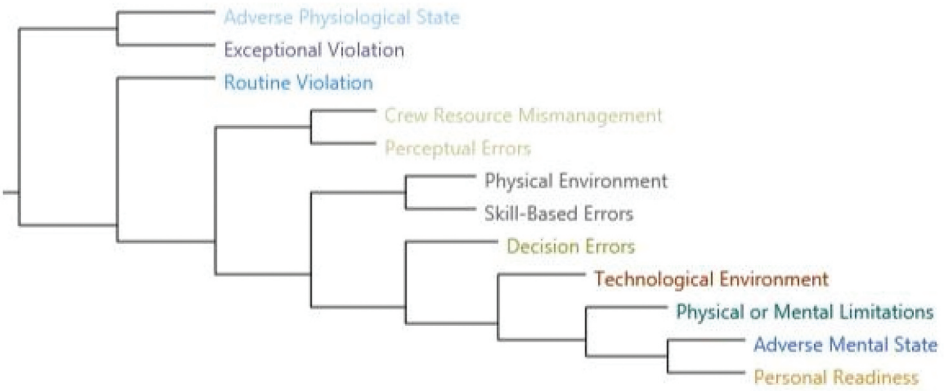


Figure 3. Nodes clustered by coding similarities. Similar nodes are represented using same color.

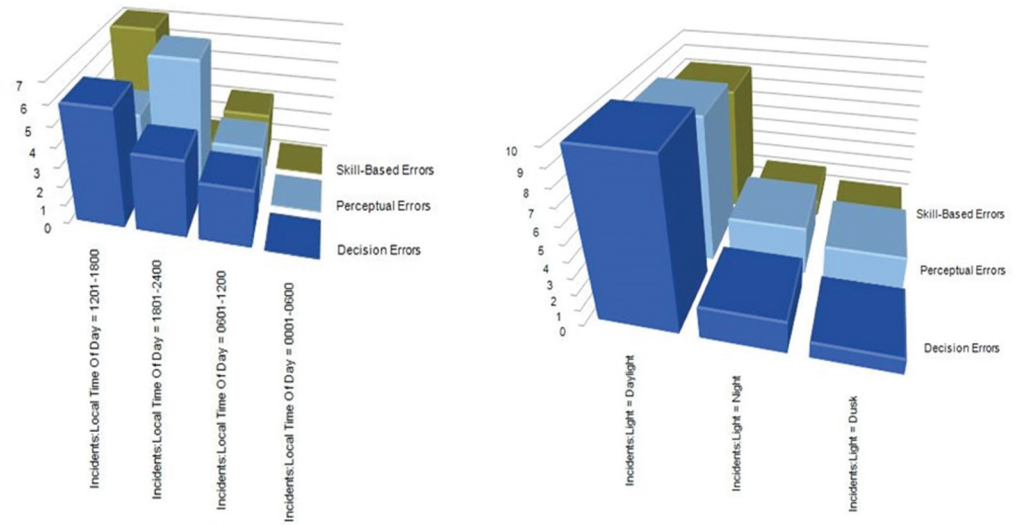


Figure 4. HFACS errors based on time of day and daylight hours.

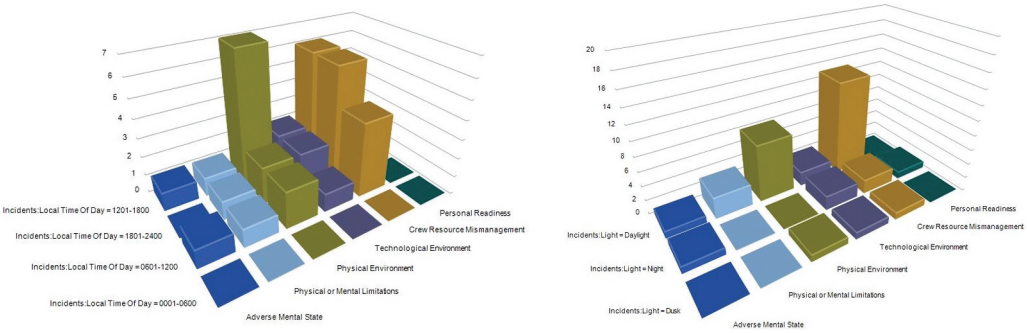


Figure 5. HFACS Preconditions based on time of day and daylight hours.

communication between pilot and air traffic control or self-reporting by the conflicting aircraft in non-towered airports. In contrast, albeit to a lesser degree, there were several contributing factors such as deviation, conflict, runway excursions, ATC and ground events that resulted in incidents with precondition of physical environment. The classification of HFACS errors to the reported contributing factors is shown in Figure 7. Again, conflict-based incidents were the primary cause of perceptual and decision-based errors, while skill-based errors although fewer in number could be attributed to different factors such as deviation, conflict, equipment, in flight or ground event.

Discussion

Lenné et al. (2008) found that personal readiness, physical and mental limitations, and adverse mental state were preconditions to skill-based or decision errors. However, an analysis of the

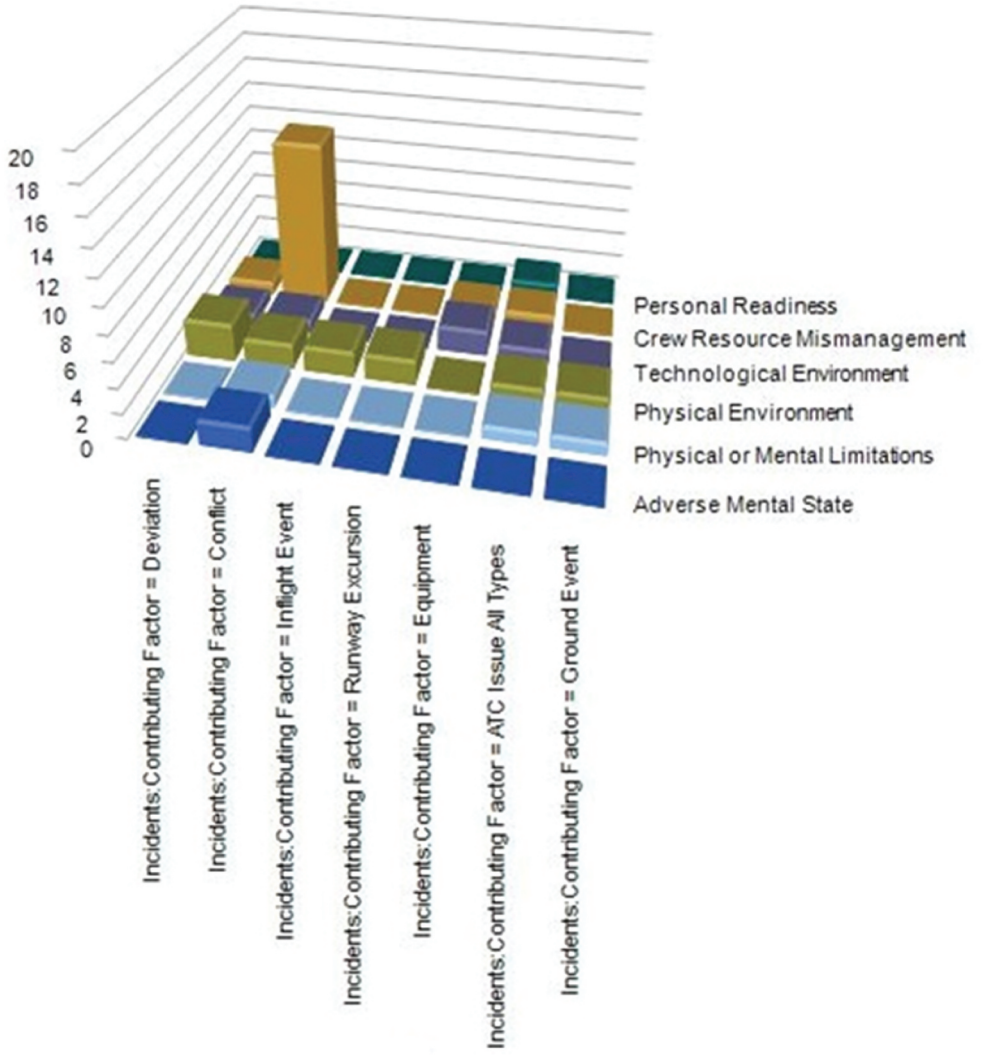


Figure 6. Classification of contributing factors to HFACS model.

ASRS reports did not find data to support any such relationship. Lenné et al. (2008) also observed that crew resource management was most often associated with a person committing a violation, whereas the ASRS data indicated perceptual errors were more closely associated with crew resource management. The differences between these findings, and the findings of Wiegmann and Shappell (2001), indicate that the self-reported ASRS incidents may have limited external validity and should not be applied to all general aviation accidents or incidents.

As already discussed, there is evidence to show that pilots might not perceive their own faults and therefore this nuance of reporting could be dramatically understated. While the researchers feel that this circumstance is not as severe as Stanford and Homan (1999) warned against in their ethics in ASRS reporting study, some concessions were made regarding the self-assessment capability of the reporters. Potential inaccuracies in reporting

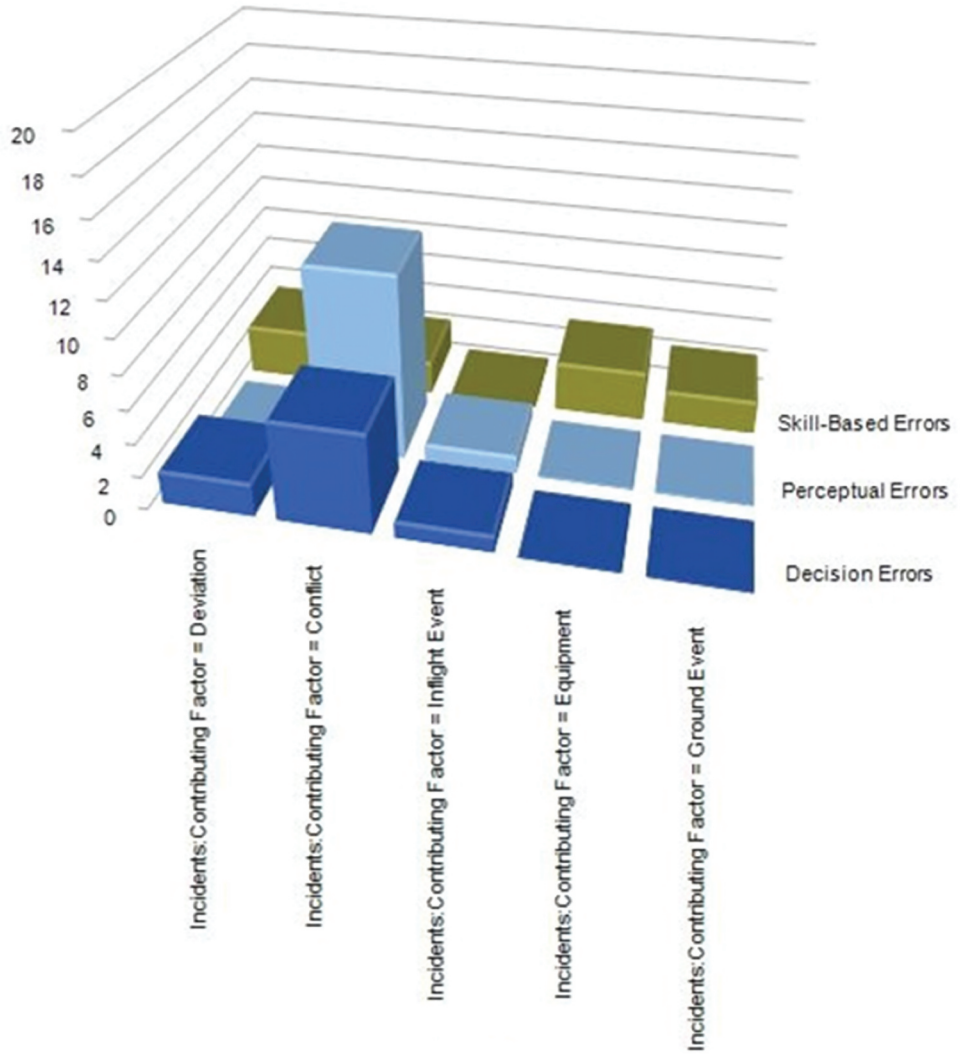


Figure 7. Classification of contributing factors to HFACS errors.

were presumed to be a matter of perception and lack of understanding, rather than any willful violation or attempt to obscure the truth.

The majority of the reports identified poor environmental conditions, perception errors from not seeing another aircraft, and crew resource management conditions due to another pilot or tower failing to properly communicate. There were no self-reported violations or adverse physiological states. There were only three adverse mental states, mental limitations, or personal readiness issues; and those incidents were inferred by the subject matter experts based upon pilot experience levels and other situational conditions such as high traffic and confusion. Because the concession was made that individuals may be less likely to report their own faults, external validity is limited, and these results should not be generalized for non-reported incidents.

The application of the HFACS framework to ASRS general aviation reports indicated most incidents were perceptual errors, followed by decision errors and then skill-based errors. The primary precondition for those errors was crew resource management, physical environment and technological environment, respectively. In general, the majority of the incidents can be categorized as a crew being unable to see, hear, or communicate with another aircraft. There were some qualitative correlations between crew resource management and perceptual errors; and between physical environment and skill-based errors.

The most frequent precondition for an unsafe act was crew resource management. Wiegmann and Shappell (2001) also found that this was the most common precondition for commercial aviation accidents. These findings indicate that there may be a relationship between general aviation and commercial aviation with regards to the preconditions for unsafe acts. However, the frequencies and percentages of unsafe acts in ASRS findings differed significantly from Wiegmann and Shappell (2001) and from Lenné et al. (2008). Because individuals might be reluctant to accurately self-report, particularly if they feel they might be at fault, these differences likely do not reveal the true relationships about the type of unsafe acts known to be associated with GA incidents.

Recommendations

For this research, the researchers examined each ASRS incident and then achieved a consensus for the primary unsafe act and precondition for that act. However, many incidents contained multiple errors and preconditions. In a future study, each researcher should individually code the incidents, using multiple HFACS categories if applicable. The researchers could then achieve consensus and compare the results of their individual coding to each other and to the consensus classification. More incidents from previous years and from other aircraft types could also be included to increase the number of usable samples in the dataset. This extra level of analysis would provide greater validation of the results.

Additional analyses should also be performed to examine and compare the HFACS frequencies for incidents in which a pilot reported an unsafe act that they committed, against incidents where they reported an unsafe act that another person committed.

Conclusions

The HFACS framework is widely used to classify the factors of military aviation incidents and accidents. Wiegmann and Shappell (2001) showed that that framework also can apply to commercial aviation accidents. This research showed that the HFACS framework also has applications for general aviation incidents as reported through the ASRS database. The primary unsafe act for the incidents analyzed was perceptual errors and the primary precondition for these unsafe acts was crew resource management. The crew resource management precondition coincides with the findings of Wiegmann and Shappell (2001) and others. This commonality indicates that general aviation incidents and accidents may have similar causes as commercial and military aviation. However, the nature of the ASRS errors did not match previous findings in commercial or military aviation. This result indicates that general aviation pilots may make different types of errors, or more likely, that they do not voluntarily report those errors for which they are at fault.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

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